

**IMPORTANT:**

This resource was developed to support a previous version of the syllabus.  
It may contain content that differs from the current syllabus.

## **HSC Marking Feedback 2017**

### **Mathematics General 2**

### **Written Examination**

#### **Question 26**

##### **General Feedback**

##### **Students should:**

- understand the term 'part thereof' and 'for each call' (part (b))
- capture/recapture using a proportion statement rather than a memorised formula (part (c))
- learn to read and interpret an area chart (part (f)).

##### **Student Strengths**

##### **In better responses, students were able to:**

- calculate the costs involved using electricity (part (a))
- use right angled trigonometry to calculate a side (part (d))
- calculate share cost, brokerage, selling price and dividend (part (e)).

#### **Question 27**

##### **General Feedback**

##### **Students should:**

- use the statistics function on a calculator to find the standard deviation (part (a) (ii))
- convert between units – megabytes and terabytes (part (b))
- calculate from a given table the future value of an annuity and the total interest earned as opposed to the interest earned per annum (part (c)).

##### **Student Strengths**

##### **In better responses, students were able to:**

- calculate the mean of a data set (part (a))
- calculate the longitude of a place from a diagram (part (d) (i))
- show a conversion from degrees in longitude to hours (part (d) (ii)).

## Question 28

### General Feedback

#### Students should:

- solve simultaneous equations and explain the context of the solution (parts (a) (ii) and (iii))
- be competent with  $P + I - R$  using more than one application (part (c))
- change the subject of a formula involving the inverse of a square root (part (d)).

### Student Strengths

#### In better responses, students were able to:

- write down sample spaces (part (b))
- read and interpret a graph (part (e))
- calculate the profit earned after costs are calculated and subtracted (part (e)).

## Question 29

### General Feedback

#### Students should:

- verify a given solution by showing calculations on how the result was obtained (part (a))
- convert  $\text{km}^2$  to  $\text{m}^2$  (part (b))
- recognise when data does not fit a normal distribution /bell curve (part (d) (ii) and (iii)).

### Student Strengths

#### In better responses, students were able to:

- calculate probability using a two-way table and finding two relevant probabilities and comparing them (part (c))
- calculate the Medicare levy, and taxable income from a given table (part (b))
- find the median from a dot plot (part (d)).

## Question 30

### General Feedback

#### Students should:

- understand the difference between the range and IQR for a data set (part (a))
- realise variation is not always linear (part (b))
- understand a bearing as opposed to an angle (part (c)).

### Student Strengths

#### In better responses, students were able to:

- find the IQR (part (a))
- apply the cosine rule to find the side length of a triangle and sine rule for an angle (part (c))
- substitute into a given formula (part (d)) and selecting the appropriate formula from the reference sheet provided to calculating the volume of a cone and area of a circle (part (e)).